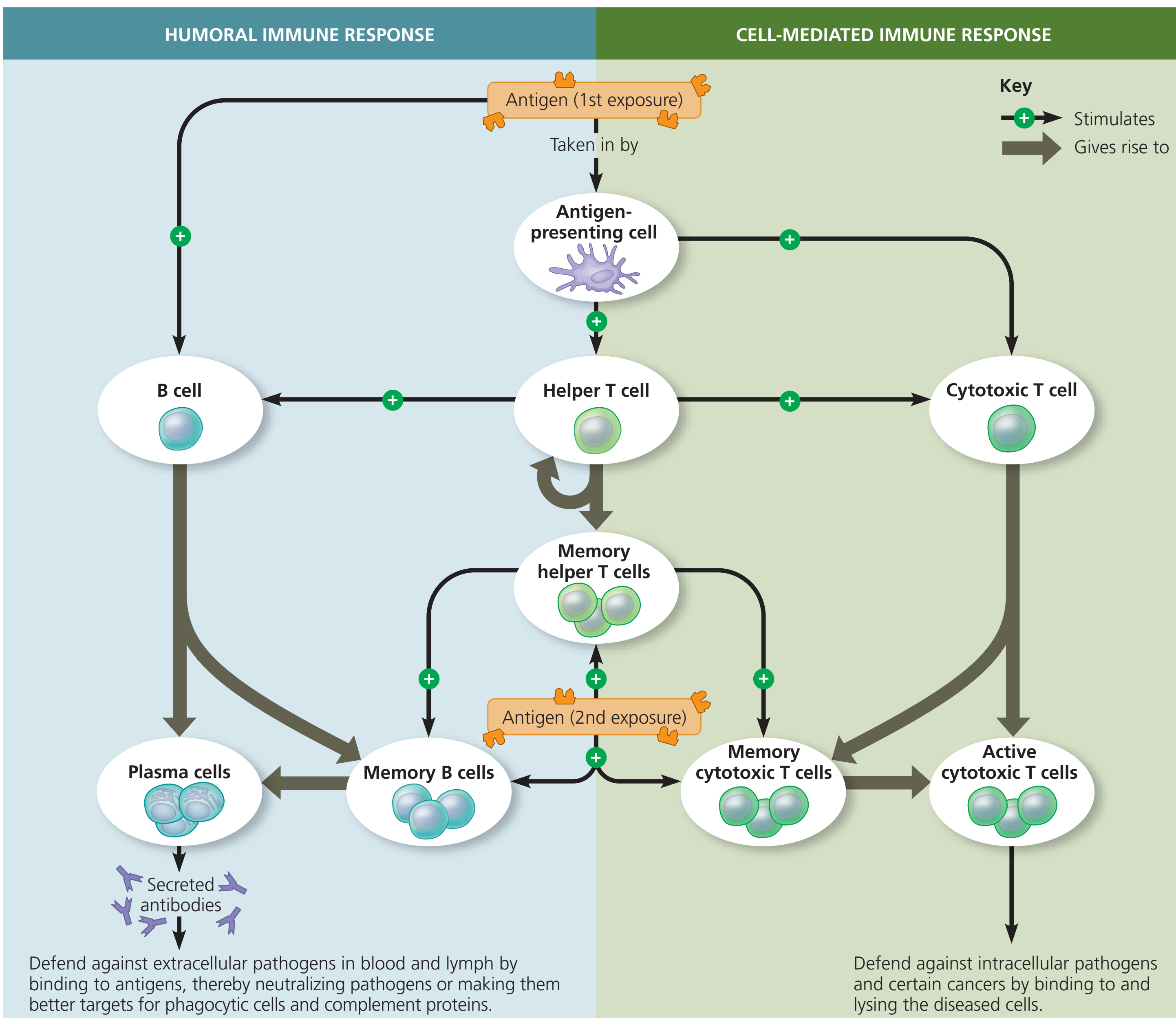
T cells. Like helper T cells, cytotoxic T cells have an accessory protein that can bind to an MHC molecule. The interaction of this accessory protein, called CD8, with a class I MHC molecule displaying antigen targets the infected cell for cytotoxic T cell activity.

The targeted destruction of an infected host cell by a cytotoxic T cell involves the secretion of proteins that disrupt membrane integrity and trigger cell death (apoptosis; see Figure 43.22). The death of the infected cell not only deprives the pathogen of a place to multiply but also exposes cell contents to circulating antibodies, which mark released antigens for disposal.

## Summary of the Humoral and Cell-Mediated Immune Responses

As noted earlier, both humoral and cell-mediated immunity can include primary as well as secondary immune responses. Memory cells of each type—helper T cell, B cell, and cytotoxic T cell—enable the secondary response. For example, when body fluids are reinfected by a pathogen encountered previously, memory B cells and memory helper T cells initiate a secondary humoral response. **Figure 43.23** summarizes adaptive immunity, reviews the events that initiate humoral and cell-mediated immune responses, highlights the difference in

▼ **Figure 43.23 An overview of the adaptive immune response.**




**VISUAL SKILLS** *Identify each arrow as representing part of the primary response or secondary response.*

**Mastering Biology BioFlix® Animation: Summary of the Adaptive Immune Response (Example: Infection by a Rhinovirus)**